

KOS IT & Admin Manual: Infrastructure, Security & Compliance

1. Governance & Architecture

The Lobed Knowledge Operating System (KOS) is a "Zero-Trust" educational architecture designed to run exclusively within the ccpsnet.net Google Workspace environment. It utilizes Google's native Apps Script engine and Gemini API, requiring no external server hosting or third-party API keys.

2. FERPA & Data Privacy Compliance

The system enforces a "Walled Garden" through the following technical constraints:

- **No Data Leakage:** All student data (JSON, TXT, MD) resides in Google Drive folders owned by the Teacher.
- **OAuth Scopes:** Scripts use restricted Google Drive and Chat scopes.
- **PII Protection:** Student identity is tied to their internal district email. No personal identifiers are transmitted outside the CCPS domain.

3. Orchestration & Performance (The Performance Layer)

To ensure system stability across the district infrastructure, the KOS utilizes:

- **CacheService (Tier 1):** Reduces Drive API load by storing session state in Google's distributed RAM for 10-minute windows.
- **UrlFetchApp (Parallel I/O):** Bypasses standard DriveApp bottlenecks by using REST-level concurrent fetches for JSON shards.
- **LockService:** Prevents race conditions and "Collision Errors" on shared files during high-concurrency periods (e.g., the start of a class period).

4. Security Guardrails: Prompt Injection & Tampering

- **Narrative Sanitization:** The Orchestrator includes a regex-based filter that scrubs student-edited STUDY_LOG.txt files for injection strings (e.g., system_note:) before ingestion by the LLM.
- **Tri-Tier Permissions:**
 - **System Files (JSON):** Set to Viewer for students to prevent manual score manipulation.
 - **Inlet Files (TXT/MD):** Set to Editor for student agency.
 - **Locked Artifacts:** Post-verification, the system automatically demotes student access to Viewer on linked Google Docs/Slides to preserve the "Record of Truth."

5. Maintenance & Self-Healing

- **Compaction Engine:** Automatically shards data to prevent file bloat and API timeouts.
- **Circuit Breaker:** Detects Drive API latency and switches the environment to "Read-Only" mode to prevent transaction failure or data corruption.
- **Transaction IDs (TX-ID):** Every data change is stamped with a unique ID, providing a forensic audit trail for all mastery updates.

6. Deployment Verification

Before approving a KOS instance, ensure the following:

1. **Script Owner:** Must be a ccpsnet.net staff account.
2. **Deployment Type:** "Web App" or "Chat App" configured to "Execute as: User" and "Access: Anyone within CCPS."
3. **Scopes:** No unauthorized third-party URLs in the appsscript.json manifest.